4-Stage Serial Data Delay Line (SISO Shift Register): RTL to GDS in SKY130

Process / library: SkyWater SKY130, `sky130_fd_sc_hd` standard cells (PDK build `c6d73a35f524070e85faff4a6a9eef49553ebc2b`, installed with volare) **Tools:** Icarus Verilog + GTKWave (simulation), Yosys (synthesis), OpenROAD 0.9.0 / OpenSTA / TritonRoute (physical design and timing, `efabless/openlane:v0.21` image), KLayout 0.28.16 (images, GDS generation, DRC), Magic 8.3.685 + netgen 1.5.185 (extraction and LVS) **Clock period:** 10 ns (100 MHz), see `constraints.sdc`

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0. Project proposal

Title and application. 4-stage serial data delay line, used to align a serial data stream with a control signal that arrives a fixed number of clock cycles earlier (typical of serial-link and sensor-interface logic).

Problem statement. When a data stream and its control or strobe signal are skewed by a known number of clock cycles, the receiving logic samples the wrong bit. Correcting this in software costs processor time and adds timing variation. A small dedicated hardware delay is cycle-exact and needs no processor.

Proposed solution. A 4-flip-flop shift register with a synchronous reset. Every bit on `serial_in` appears on `serial_out` exactly 4 clock edges later.

Main functionality. Fixed 4-cycle serial delay, cleared by reset.

Signal	Direction	Description
`clk`	input	Clock, rising-edge active
`rst`	input	Synchronous reset, active high (clears all four stages)
`serial_in`	input	Serial data in
`serial_out`	output	`serial_in` delayed by 4 clock edges

Sequential logic. A 4-bit shift register of D flip-flops (`shift_reg[3:0]`). The output is the last stage.

Test cases (`tb/siso_shift_register_tb_sc.v`): reset with `serial_in` held high; directed pattern `10110010`; all ones then all zeros; alternating `1010...`; latency (a single 1 appears after exactly 4 edges); reset in the middle of a stream; reset held for 6 cycles; 300 random cycles with random resets.

Feasibility. 8 functional cells (4 flip-flops, 4 NOR2B gates) plus clock buffers and physical cells, in a 38.7 µm × 38.7 µm die. The design was taken through simulation, synthesis, place and route, timing in three corners, GDS generation, DRC and LVS.

> My project is a **4-stage serial data delay line** that solves **the alignment of a serial data stream with a control signal that arrives a fixed 4 clock cycles earlier** by taking **a serial bit stream, a clock and a synchronous reset** as inputs and producing **the same bit stream delayed by exactly four clock cycles** as output. It uses **a 4-bit shift register built from D flip-flops** as its main

sequential logic.

The design is deliberately small. The aim is a complete, verified path from application to GDS.

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1. Design description

```
always @(posedge clk) begin
    if (rst) shift_reg <= 4'b0000;
    else     shift_reg <= {shift_reg[2:0], serial_in};
end
assign serial_out = shift_reg[3];
```

After synthesis the logic is 4 `dfxtp_1` flip-flops and 4 `nor2b_1` gates. The NOR2B cells implement the synchronous reset, one per stage (each stage's D input is the previous stage ANDed with NOT `rst`).

Cells in the final layout (`pnr/06_droute_fill_fix3.def`, 37 components):

Cell	Count	Origin
`dfxtp_1`	4	Synthesis
`nor2b_1`	4	Synthesis
`clkbuf_16`	3	Clock tree synthesis
`decap_4`	7	Physical flow (decoupling)
`tapvpwrvgnd_1`	4	Physical flow (well taps)
fill cells	15	Physical flow

Die size: 38.715 µm × 38.715 µm (about 1499 µm²).

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2. Flow summary

#	Step	Tool	Result
1	RTL and functional verification	Icarus Verilog	Pass (Section 3)
2	Logic synthesis and SKY130 mapping	Yosys	Done
3	Preliminary multi-corner STA	OpenSTA	Positive slack
4	Floorplan, power planning, pin placement	OpenROAD	Done
5	Global placement, pre-route STA	OpenROAD / OpenSTA	Setup 8.72 ns, hold 0.43 ns
6	Clock tree synthesis	TritonCTS	Setup 8.49 ns, hold 0.43 ns

#	Step	Tool	Result
7	Global and detailed routing	OpenROAD / TritonRoute	Done
8	Post-routing STA, 16 corners	OpenSTA	Section 5.3
8b	Parasitic extraction	OpenLane SPEF extractor (OpenRCX crashed)	Done, 16-corner STA with SPEF (Section 5.4)
9	Fillers, connectivity repair	scripts + Magic/ KLayout probes	Shorts found and repaired (Section 6)
10	GDS generation	KLayout	`gds/siso_shift_register_fix3_kl.gds`
11	DRC	KLayout, SkyWater `sky130A_mr.drc`	0 violations (Section 7)
12	LVS	Magic extraction + netgen	Circuits match (Section 7)
13	Gate-level simulation of the post-route netlist	Icarus Verilog	Pass (Section 3)

Not included, on purpose: design for test (scan chains). A 4-flip-flop design does not need them.

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3. Functional verification

`tb/siso_shift_register_tb_sc.v` has a reference model, compares `serial_out` with it on every clock cycle, and prints PASS or FAIL. It also contains a latency test that does not use the model.

Run	Checks	Errors	Result
RTL	416	0	PASS
Mapped gate-level netlist (`synth/ siso_shift_register_sky130.v`, behavioral cell models)	416	0	PASS
Post-route netlist with clock buffers (`netlist/ siso_shift_register_postroute.v`, with the `serial_out` link and a wire declaration added in a temporary copy, behavioral cell models)	416	0	PASS
RTL with a deliberate bug (output taken from `shift_reg[2]`)	416	163	FAIL (bug detected)

All simulations are functional and zero-delay. Timing is covered by STA.

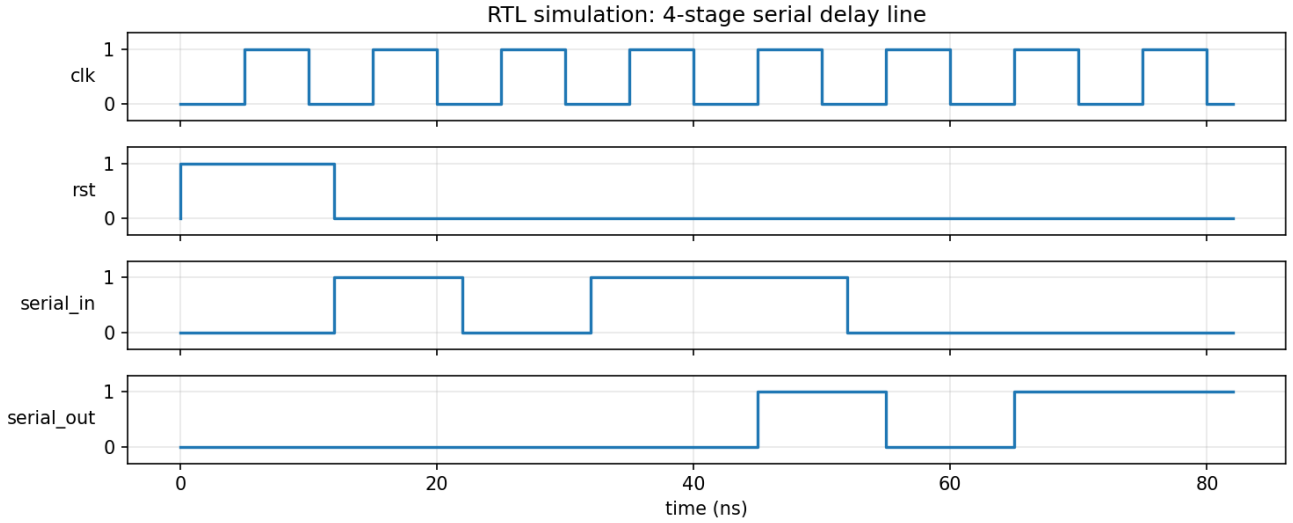


Figure 1: RTL simulation waveform (`clk`, `rst`, `serial_in`, `serial_out`).

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4. Physical design results

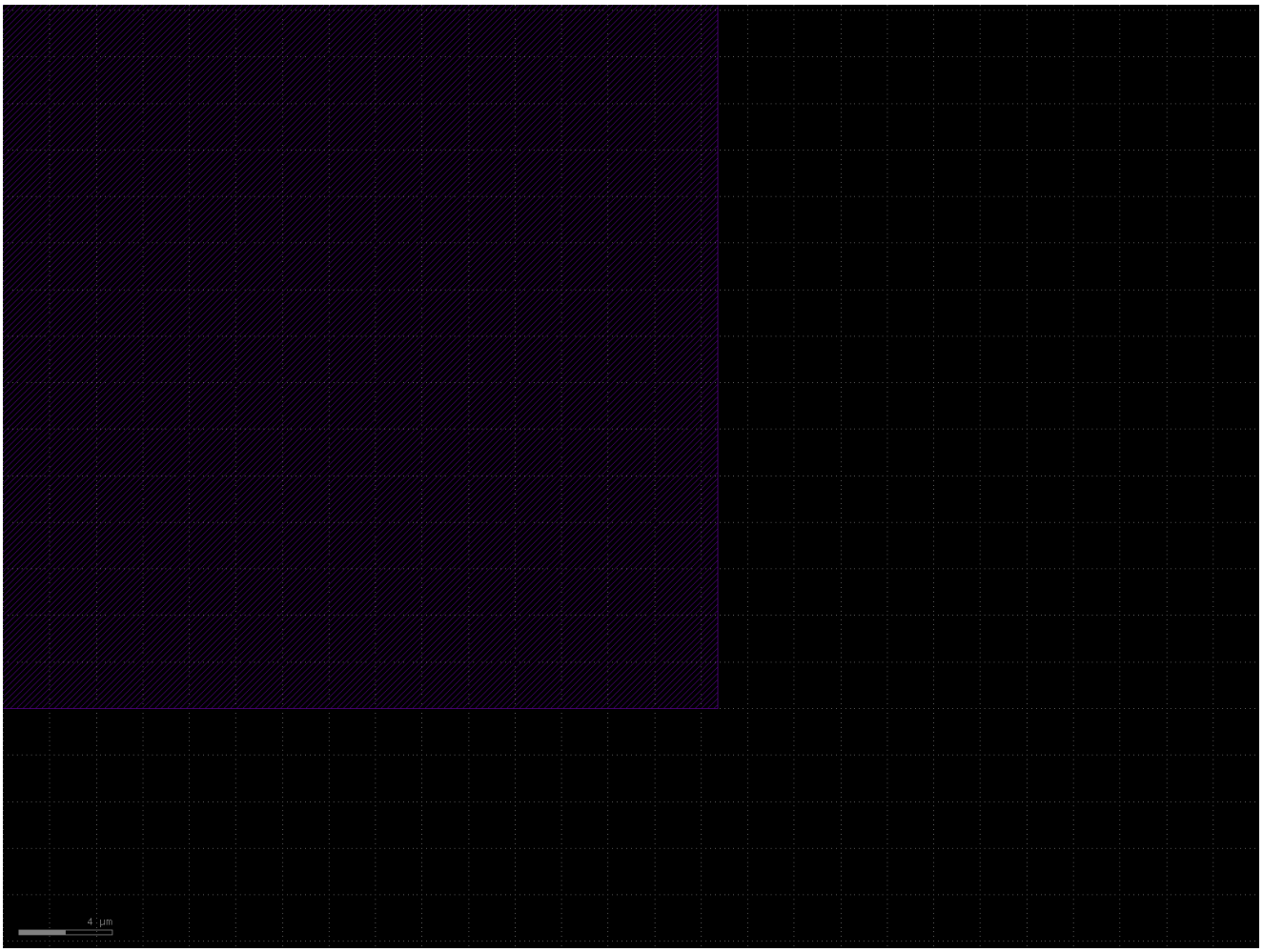


Figure 2: Floorplan.

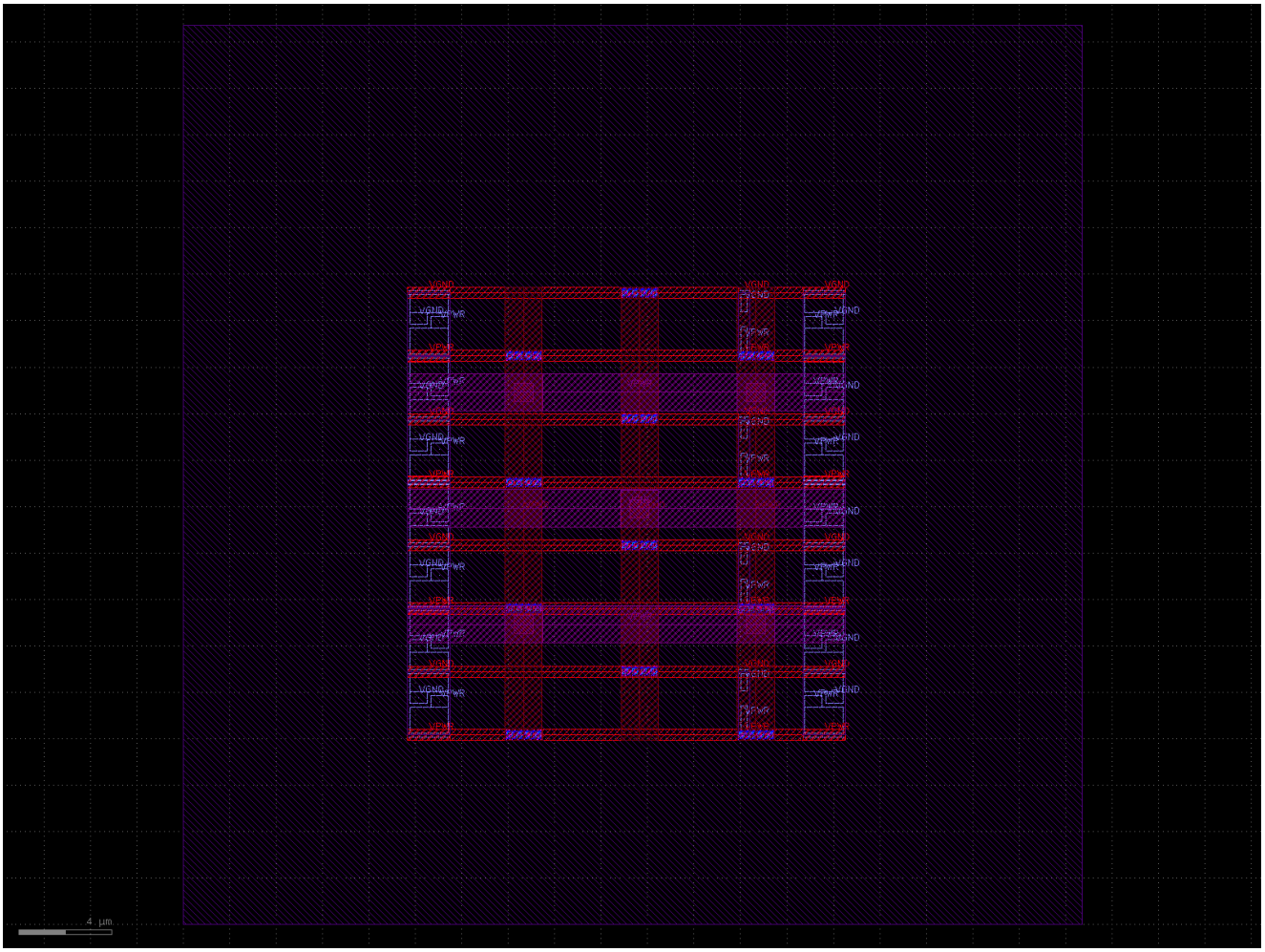


Figure 3: Power distribution network.

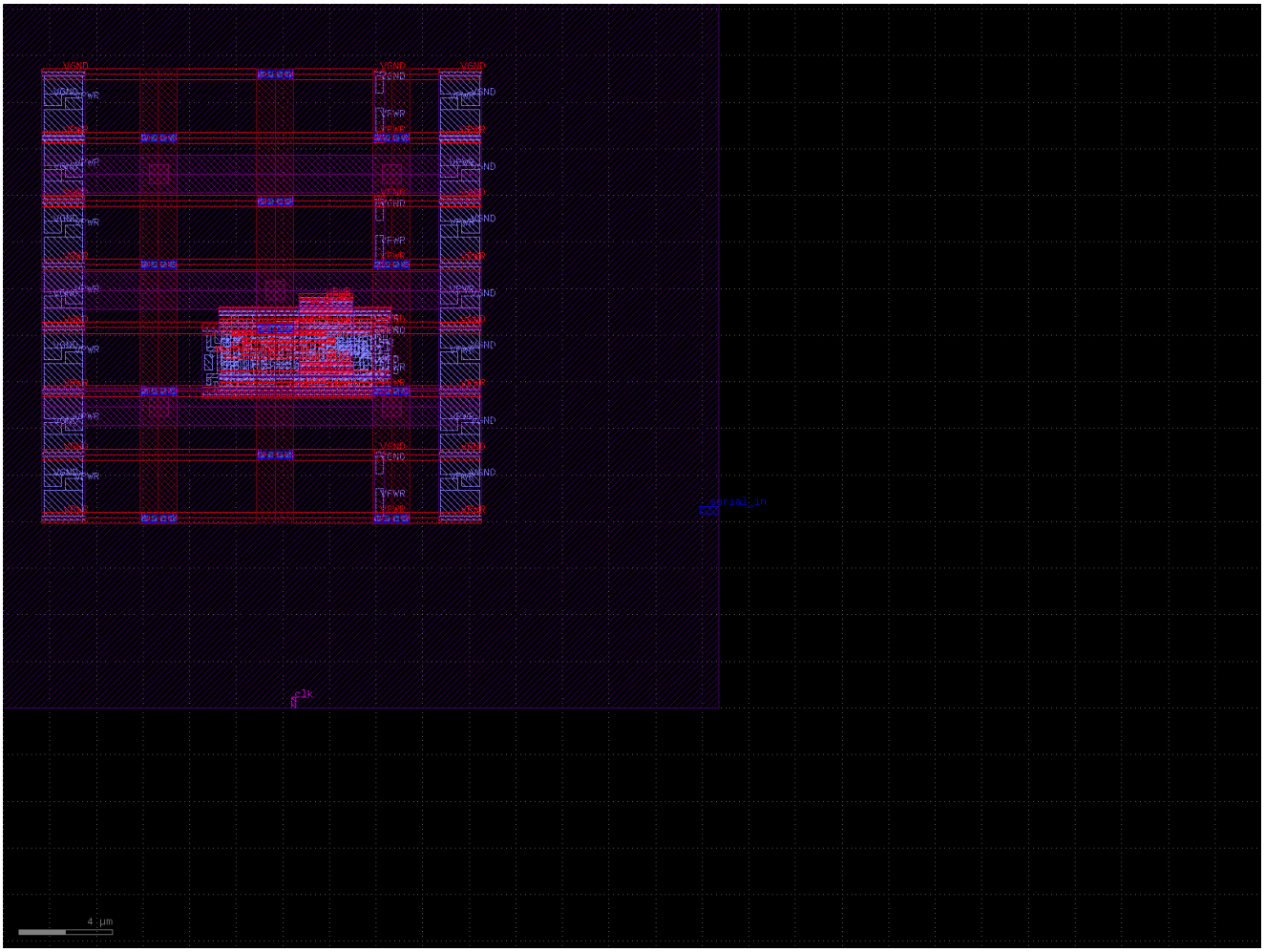


Figure 4: Pin placement.

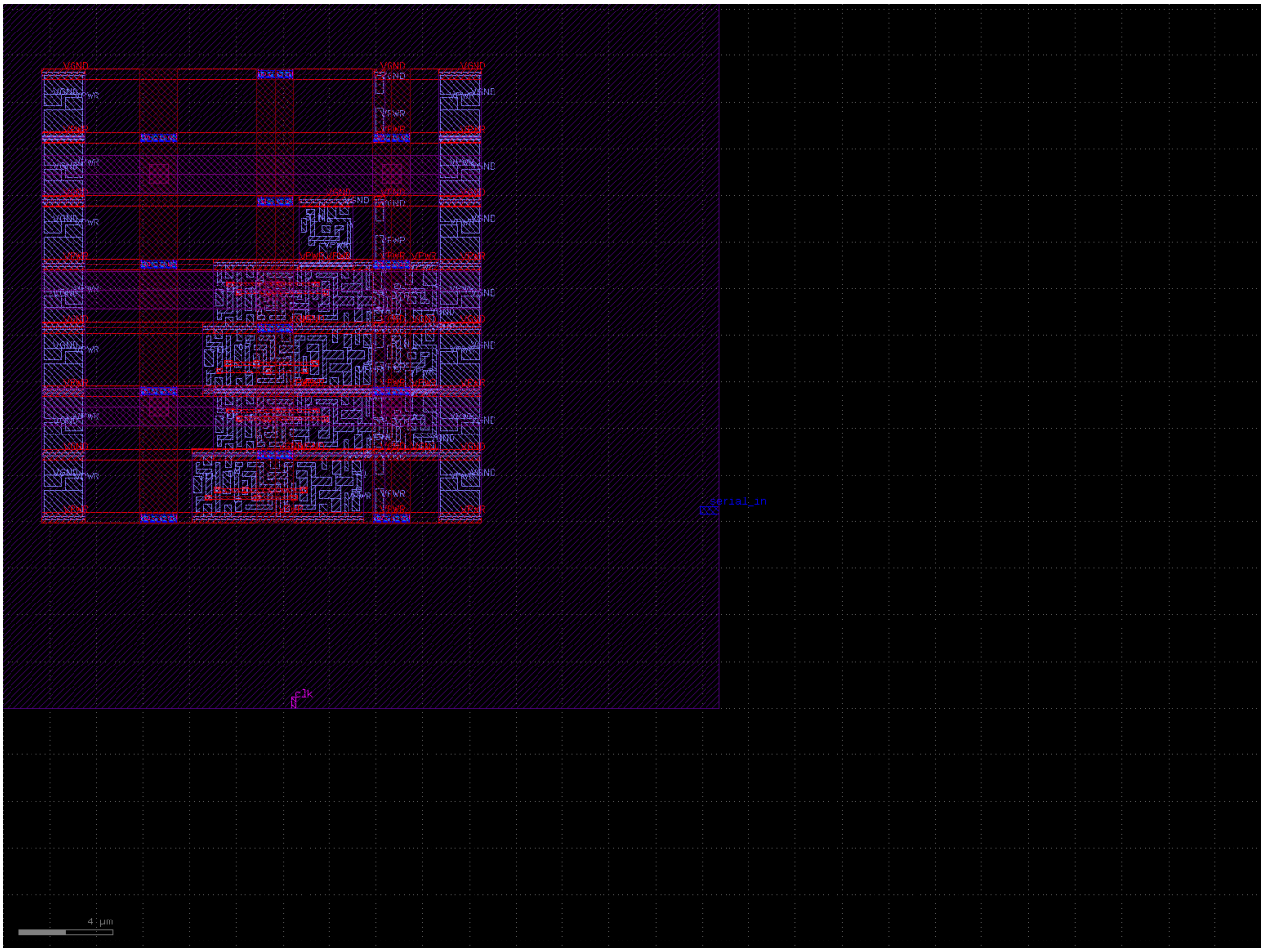


Figure 5: Global placement.

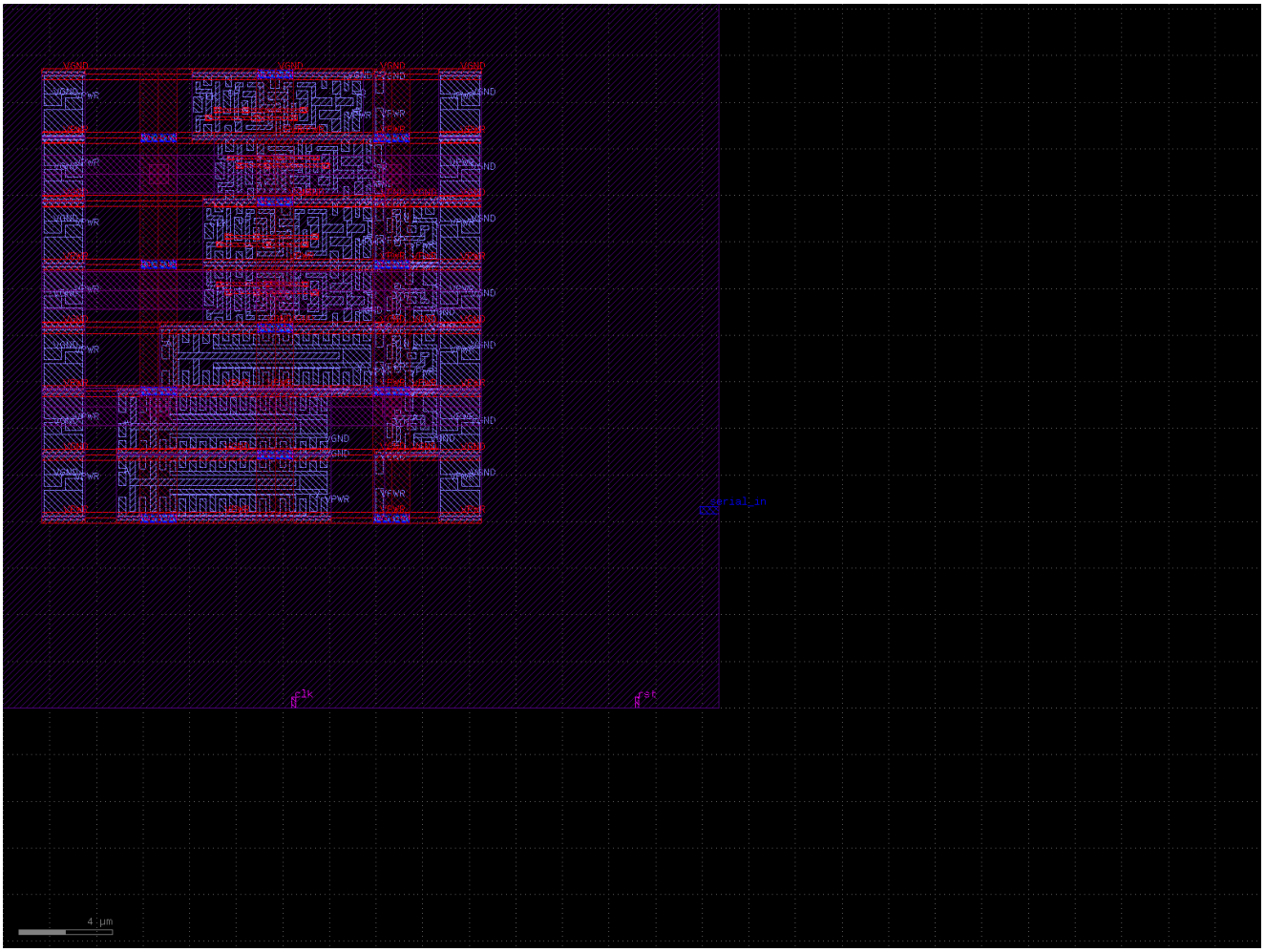


Figure 6: After clock tree synthesis (3 `clkbuf_16` buffers).

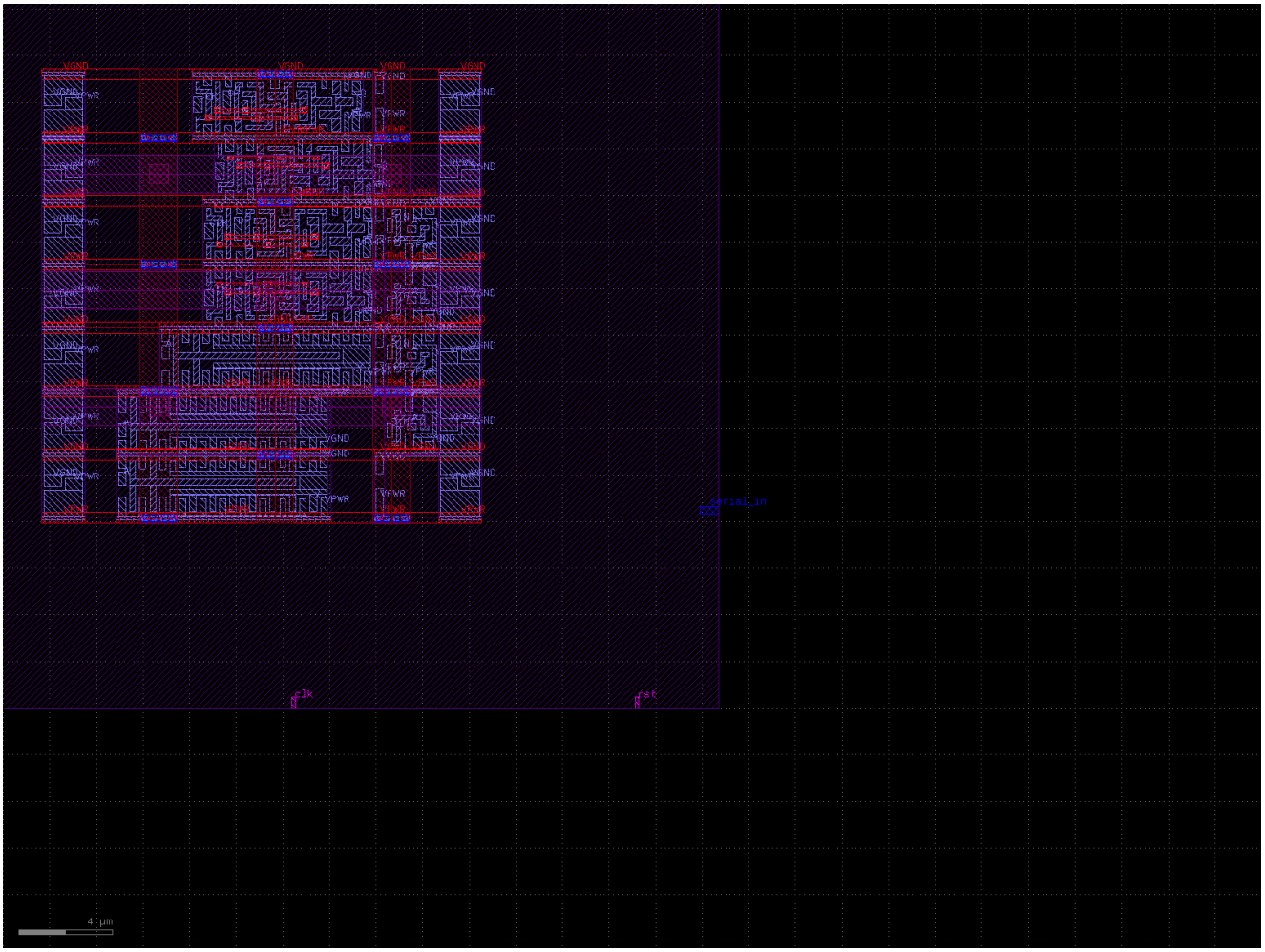


Figure 7: Global routing.

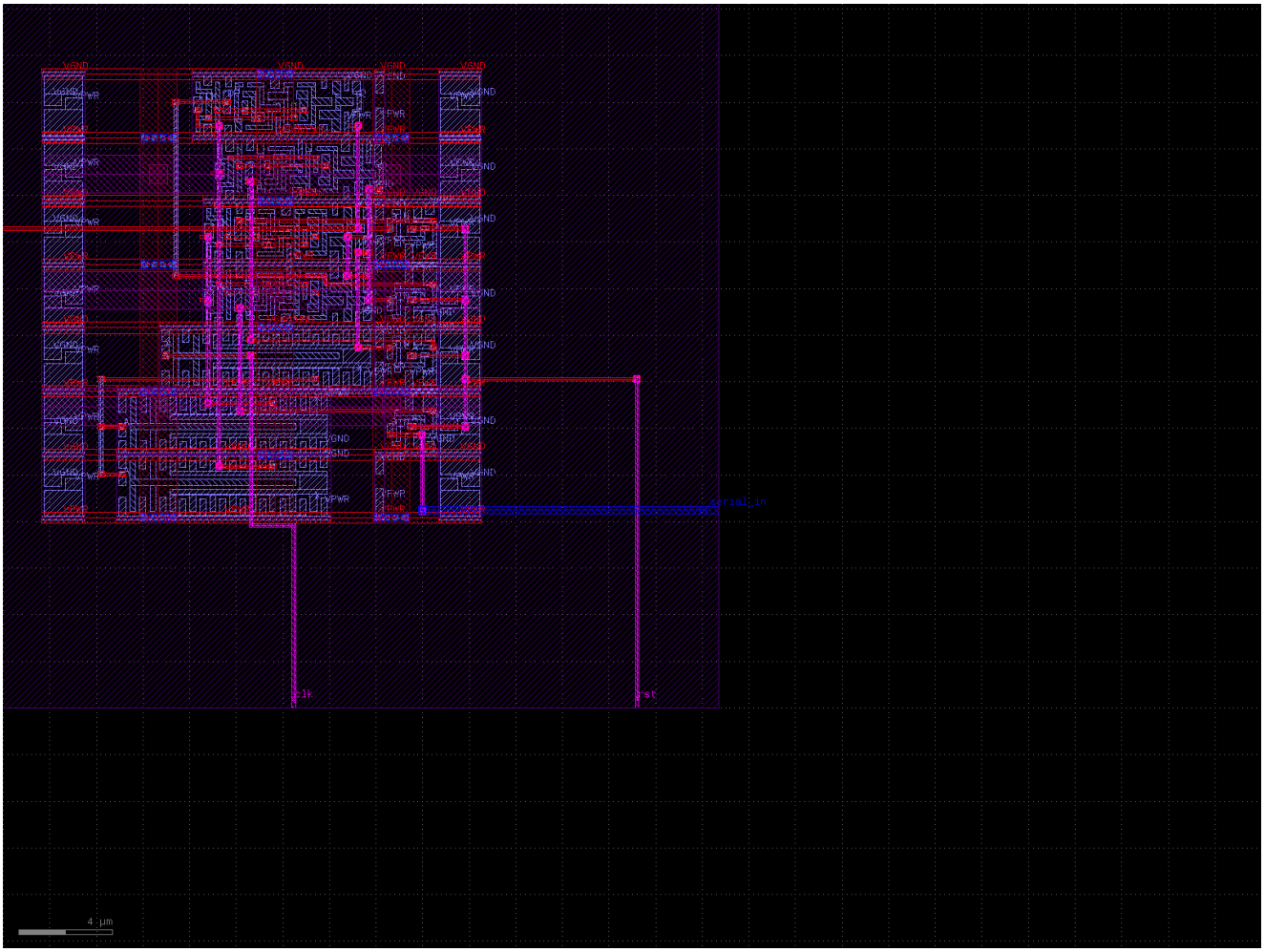


Figure 8: Detailed routing.

Figures 2 to 8 were rendered from the DEF files of each stage, before the repairs described in Section 6.

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5. Timing results

Stage	Setup slack	Hold slack
Pre-routing	8.72 ns	0.43 ns
After CTS	8.49 ns	0.43 ns

Post-routing STA, 10 ns clock, propagated through the CTS buffers. Wire parasitics are estimated from placement (`estimate_parasitics -placement`, `set_wire_rc -layer met2`); a SPEF extracted with OpenLane's SPEF extractor is used for the 16-corner timing in Section 5.4.

Corner	Library	Setup slack	Hold slack	Result
SS	`ss_100C_1v40`	8.38 ns	1.27 ns	MET
TT	`tt_025C_1v80`	9.49 ns	0.43 ns	MET
FF	`ff_n40C_1v56`	9.50 ns	0.42 ns	MET

Setup is tightest in the slow corner and hold is tightest in the fast corner. The critical path is one flip-flop-to-flip-flop stage with about 0.4 ns of data delay against a 10 ns period.

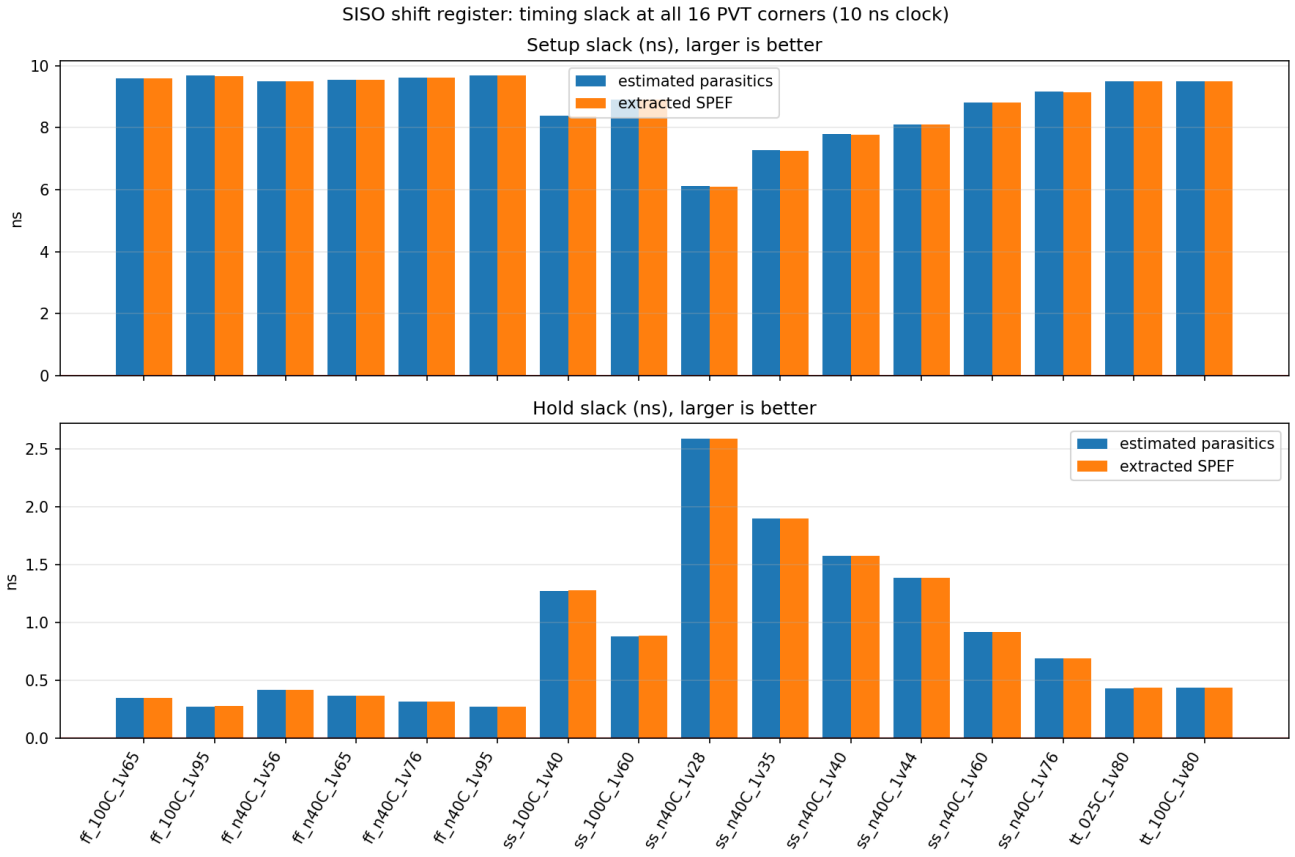
These numbers were computed on `pnr/06_droute.def`. The final layout (`pnr/06_droute_fill_fix3.def`) has an identical `NETS` section, so the signal routing is unchanged. The only cells removed were decoupling and filler cells, which have no timing function.

5.3 All 16 PVT corners (final layout)

STA was repeated on the final layout (`pnr/06_droute_fill_fix3.def`) with every liberty file shipped for `sky130_fd_sc_hd` (16 corners, named process_temperature_voltage), 10 ns clock, propagated clock, parasitics estimated from placement.

Corner	Setup slack (ns)	Hold slack (ns)	Result
ff_100C_1v65	9.59	0.35	MET
ff_100C_1v95	9.68	0.27	MET
ff_n40C_1v56	9.50	0.42	MET
ff_n40C_1v65	9.56	0.37	MET
ff_n40C_1v76	9.62	0.32	MET
ff_n40C_1v95	9.68	0.27	MET
ss_100C_1v40	8.38	1.27	MET
ss_100C_1v60	8.92	0.88	MET
ss_n40C_1v28	6.12	2.59	MET
ss_n40C_1v35	7.27	1.90	MET
ss_n40C_1v40	7.79	1.58	MET
ss_n40C_1v44	8.11	1.39	MET
ss_n40C_1v60	8.82	0.92	MET
ss_n40C_1v76	9.16	0.69	MET
tt_025C_1v80	9.49	0.43	MET
tt_100C_1v80	9.50	0.44	MET

Worst setup slack: 6.12 ns (ss_n40C_1v28). Worst hold slack: 0.27 ns (ff_100C_1v95). All 16 corners meet setup and hold.



Setup and hold slack at all 16 corners, with estimated parasitics and with the extracted SPEF.

5.4 Parasitic extraction and post-extraction STA

OpenRCX (`extract_parasitics` in OpenROAD 0.9.0) was tried first, with the extraction rules shipped with the PDK, on the final layout and on the original routed layout. Every attempt that was run ended with the exit code shown below (139 is a segmentation fault), so no SPEF file was written.

Layout DEF	Rules file type	Corner	Exit code	SPEF written
`06_droute_fill_fix3`	spef_extractor	nom	139	no
`06_droute_fill_fix3`	calibre	nom	139	no
`06_droute`	spef_extractor	nom	139	no
`06_droute`	calibre	nom	139	no

The parasitics were then extracted with the SPEF extractor that comes with OpenLane (`scripts/spef_extractor`, PI wire model, edge capacitance factor 1.0) from `pnr/06_droute_fill_fix3.def` and the merged LEF. It wrote `parasitics/final.spef` (14 nets, units ns / pF / ohm). This tool has a single corner, so the same SPEF was used for all 16 corners.

Corner	Setup slack (ns)	Hold slack (ns)	Result
ff_100C_1v65	9.59	0.35	MET
ff_100C_1v95	9.67	0.28	MET
ff_n40C_1v56	9.49	0.42	MET

Corner	Setup slack (ns)	Hold slack (ns)	Result
ff_n40C_1v65	9.56	0.37	MET
ff_n40C_1v76	9.61	0.32	MET
ff_n40C_1v95	9.68	0.27	MET
ss_100C_1v40	8.37	1.28	MET
ss_100C_1v60	8.92	0.89	MET
ss_n40C_1v28	6.10	2.59	MET
ss_n40C_1v35	7.26	1.90	MET
ss_n40C_1v40	7.78	1.58	MET
ss_n40C_1v44	8.10	1.39	MET
ss_n40C_1v60	8.81	0.92	MET
ss_n40C_1v76	9.15	0.69	MET
tt_025C_1v80	9.49	0.44	MET
tt_100C_1v80	9.49	0.44	MET

Worst setup slack: 6.10 ns (ss_n40C_1v28). Worst hold slack: 0.27 ns (ff_n40C_1v95). All 16 corners meet setup and hold. Compared with Section 5.3, the setup slack changes by at most 0.02 ns and the hold slack by at most 0.01 ns.

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6. Connectivity problem found and repaired

Finding. After the layout was rebuilt with fillers and decaps, the extracted connectivity did not match the netlist. Metal-connectivity probes on the GDS showed that the `VPWR` rail, the `VGND` rail and two signal nets (`clknet_0_clk` and `_03_`) were one electrical net. DRC did not report it, because overlapping metal of different nets simply merges into one polygon.

Cause. The fill cells had been inserted after routing. The router had run long `li1` wires through the then-empty gaps (the clock wire at x ≈ 12.2 µm and the `_03_` wire at x ≈ 15.4 µm, each crossing several row boundaries). The fill cells' power strips landed on those wires. Putting the seven fills back one at a time into the repaired layout reproduced the short each time.

Second problem. The `VPWR` and `VGND` pins in the DEF were not on the power stripes (one at x = 0.4 µm, y = 30.26 µm on met3 and one at x = 17.71 µm, y = 38.47 µm on met2), so the layout ports were not connected to the grid. They were moved onto the met4 stripes at (14.66, 20.0) µm and (19.66, 20.0) µm.

Repair. The seven fills that sat on the wires were removed. That left the seven left-column decap cells with floating n-wells, so those decaps and six neighbouring fills were removed as well. All of these cells are decoupling or filler cells with no logic. The signal routing was not touched.

Lesson. The proper flow places fillers and decaps before routing, so that the router avoids them. This repair is a workaround.

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7. Physical verification

7.1 DRC

SkyWater KLayout deck `sky130A_mr.drc` (release 2024.2.11_01.09), with the front-end (`feol`), back-end (`beol`) and off-grid (`offgrid`) groups enabled.

Layout / GDS	Violations
Layout with fillers added after routing, before the repair (KLayout-built)	6 `nwell.2a` before the fillers, 0 after (this layout still contained the shorts of Section 6)
Final layout, KLayout-built GDS (`gds/siso_shift_register_fix3_kl.gds`)	**0** (confirmed in two runs)
Final layout, Magic-built GDS (`lvs_magic/siso_magic_fix3.gds`)	2 `nsdm.2`, at x ≈ 24.4 µm, y ≈ 21.6 µm

`nsdm.2` is the minimum n+ implant (nsdm) width rule in the periphery (0.38 µm); the two marked pieces are about 0.31 µm wide. The same DEF gives no violations when KLayout writes the GDS. The raw `nsdm` shapes around the two Magic violations are identical in both files, so the rule must be reacting to a layer derived from other shapes, and the cause was not determined. The KLayout-built GDS is the final deliverable. Density, seal-ring and floating-metal checks were not run.

7.2 LVS

Layout netlists were extracted with Magic (8.3.685) from the GDS and compared with netgen (1.5.185) using the PDK `sky130A_setup.tcl`.

GDS	Devices (layout / reference)	Nets	Result
Magic-built (`siso_magic_fix3.gds`)	12 / 12	16 / 16	Circuits match uniquely
KLayout-built (`siso_shift_register_fix3_kl.gds`)	12 / 12	16 / 16	Circuits match uniquely

The four library cells compared inside each run also match against the PDK spice.

The final GDS (KLayout-built) therefore has **DRC 0 violations and an LVS match**.

How the reference netlist was made: a script converts the OpenROAD post-route Verilog into SPICE, with the cell pin order and the body connections (`VPB` to `VPWR`, `VNB` to `VGND`) taken from the PDK spice. The `serial_out` port is linked by hand to the output of flip-flop `_11_`, because the post-route Verilog does not connect them. The seven removed decaps are also removed from the reference. Tap and fill cells have no devices.

KLayout's own LVS deck was tried first. It matched a single library cell once the body pin was removed and `convert_subckts=true scale=true` were set, but it was not used for the chip-level result.

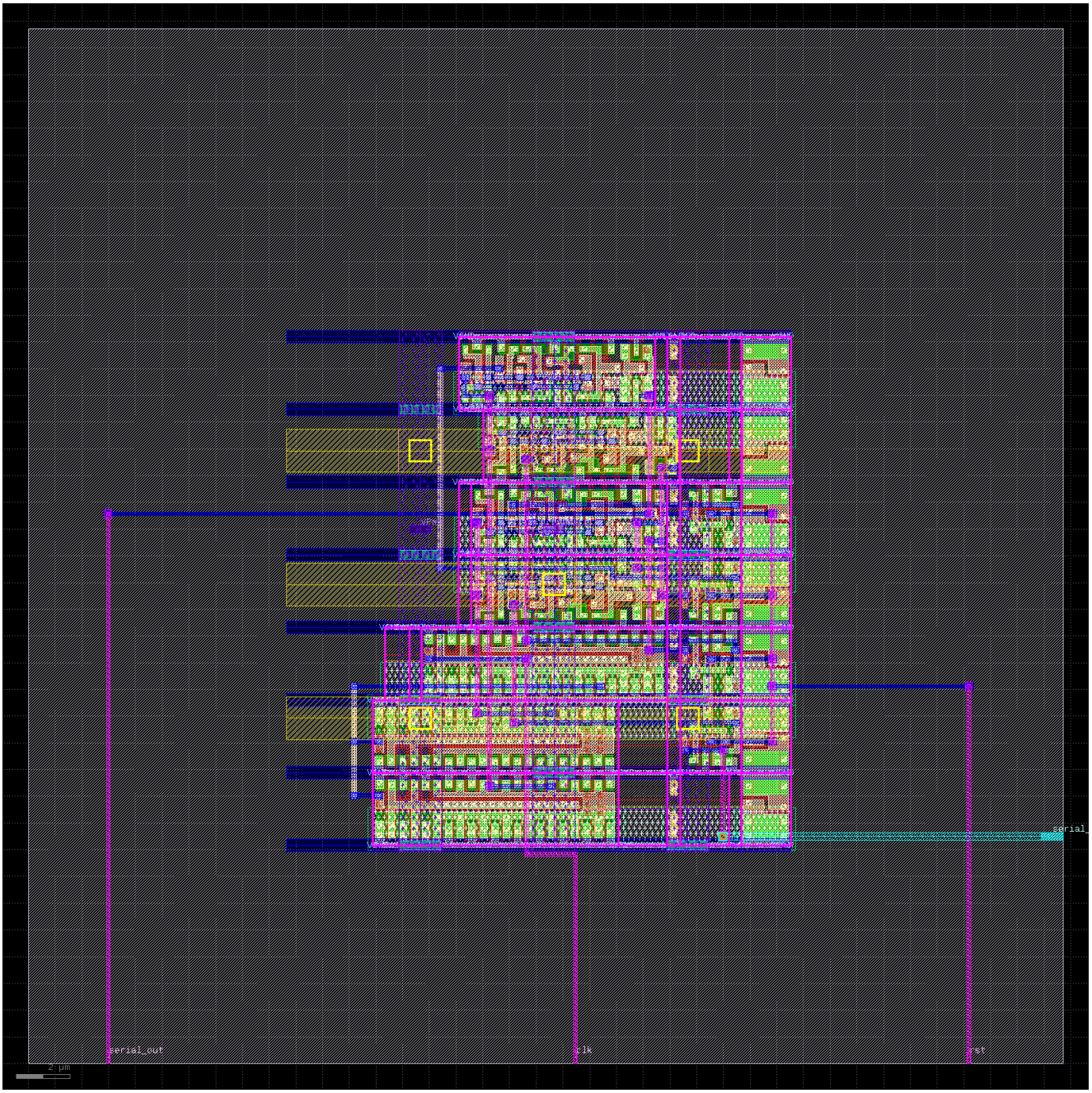


Figure 9: Final GDS layout (`gds/siso_shift_register_fix3_kl.gds`) in KLayout.

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8. Limitations

- The final layout was hand-edited from the OpenROAD output (fillers, decaps, power pins). The proper flow would place fillers before routing and generate the pins correctly.
- The LVS reference netlist was built by script and patched by hand (`serial_out` link, removed decaps). The layout and the reference are consistent with each other, but this is not a foundry sign-off flow.
- DRC covered the front-end, metal and off-grid rule groups. Density, seal-ring and floating-metal checks were not run. The Magic-built GDS shows 2 `nsdm.2` violations whose cause was not determined.
- The post-extraction timing uses OpenLane's SPEF extractor (single corner, PI wire model), not a signoff extractor, and the same SPEF is used for all 16 corners. OpenRCX could not be used because it crashes in the installed OpenROAD 0.9.0 (Section 5.4).
- No design for test, on purpose.

- The design is very small, so timing margins are large and the results are not representative of a larger design.

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9. Files

Item	Path
RTL	`final/rtl/siso_shift_register.v`
Testbenches	`final/tb/siso_shift_register_tb_sc.v` (self-checking), `final/tb/siso_shift_register_tb.v`
Constraints	`final/constraints.sdc`
Mapped and post-route netlists	`final/netlist/`
Routed and final layouts	`final/layout/06_droute.def`, `final/layout/06_droute_fill_fix3.def`
Final GDS	`final/gds/siso_shift_register_fix3_kl.gds`
Magic-built GDS	`final/gds/siso_magic_fix3.gds`
STA, DRC and LVS reports	`final/reports/`
Scripts	`final/scripts/`
Pictures	`final/images/`